Part A – Setup

Q1. Import the required Python libraries for this session:

- pandas
- numpy
- MinMaxScaler, StandardScaler from sklearn.preprocessing

Q2. Create a sample dataset named df with the following columns:

- Student_ID
- Hours_Studied
- Attendance
- Assignment_Score

and 5 sample records.
(Use your own numeric values or the ones shown below)

Student_ID	Hours_Studied	Attendance	Assignment_Score
1	4	80	75
2	6	90	85
3	5	85	78
4	7	95	92
5	3	70	65

Print the dataset.

Part B – Normalization

Q3. In your own words (Markdown cell), explain what Normalization means and when it is used.

Q4. Apply Min-Max Normalization using MinMaxScaler to scale Hours_Studied and Assignment_Score columns between 0 and 1.

Display the normalized output.

Q5. Add the normalized columns (Hours_Studied_Norm, Assignment_Score_Norm) back into the main DataFrame.

Print the updated DataFrame.

Part C – Standardization

Q6. In your own words (Markdown cell), explain what Standardization means and why it's used in analytics.

Q7. Apply StandardScaler to Attendance and Assignment_Score columns.

Create new columns named Attendance_Std and Assignment_Score_Std and print the DataFrame.

Q8. Compare the normalized and standardized results.

- What is the difference between the value ranges?
- Which one would be better for algorithms sensitive to large values?

(Write your answer in a Markdown cell.)

Part D – Reshaping with Pivot Tables

Q9. Create a new column called Attendance_Group:

- Label "High" if Attendance ≥ 85,
- Label "Low" otherwise.

Q10. Create a pivot table that shows the average Assignment_Score for each Attendance_Group.

Display the result.

Part E – Visualization

Q11. Create a line plot comparing the normalized and standardized Assignment_Score values.

Label your axes and include a legend.

Part F – Reflection

Q12. In 2–3 sentences, write your observations on:

- How normalization and standardization change the data.
- Why these techniques are important before applying machine learning models.